Microprocessor & Computer Architecture (μpCA)

UE24CS251B

Session - 9

Microprocessor & Computer Architecture (μpCA)

UE24CS251B

PSR & SWAP Instructions

PSR Instructions

Move to Register from Status Register (MRS): Read from either CPSR or SPSR

Example:

MRS R0, CPSR

MRS R1, SPSR

Move to Status Register from Register (MSR):

Example:

MSR CPSR_field, R0

MSR SPSR_field, R1

_field

_c: Control Field (0:7)

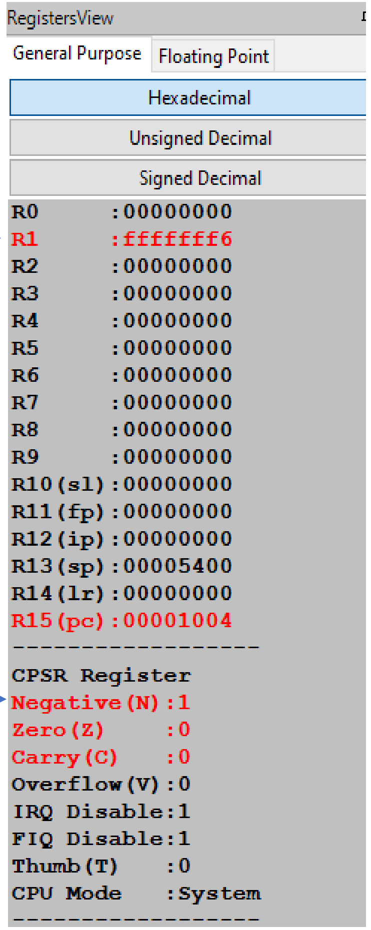
_f: Flag Field(24:31)

_x: Extension (8:15)

_s:Status (16:23)

PSR Instructions

```
.text
MOVS R1, #-10
MRS R0, CPSR                 ; Take a
copy of the CPSR.
AND R0, R0, #0000        ; Clear the
mode bits.
MSR CPSR_F, R0             ; Write back
the modified CPSR
.end
```



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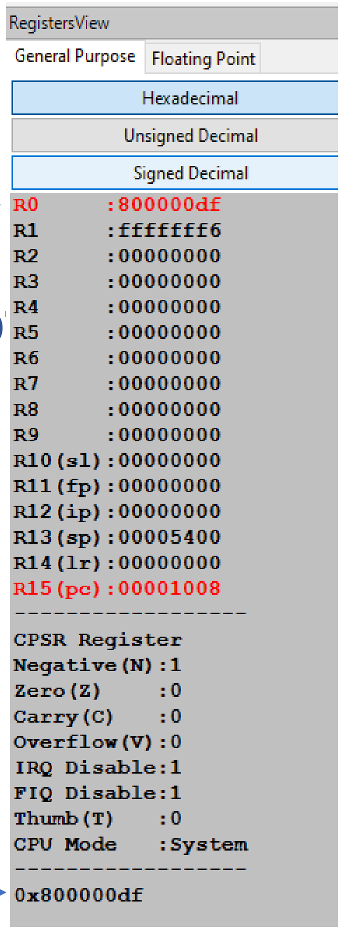
```



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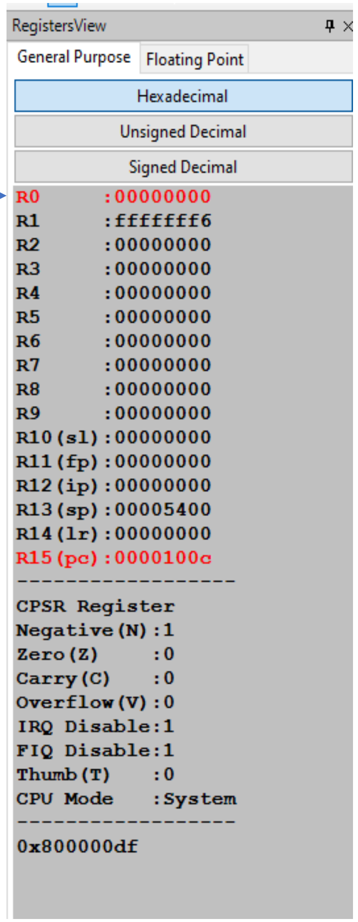

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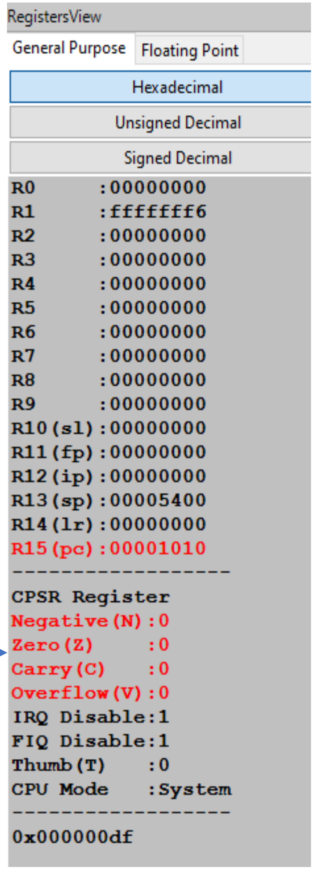
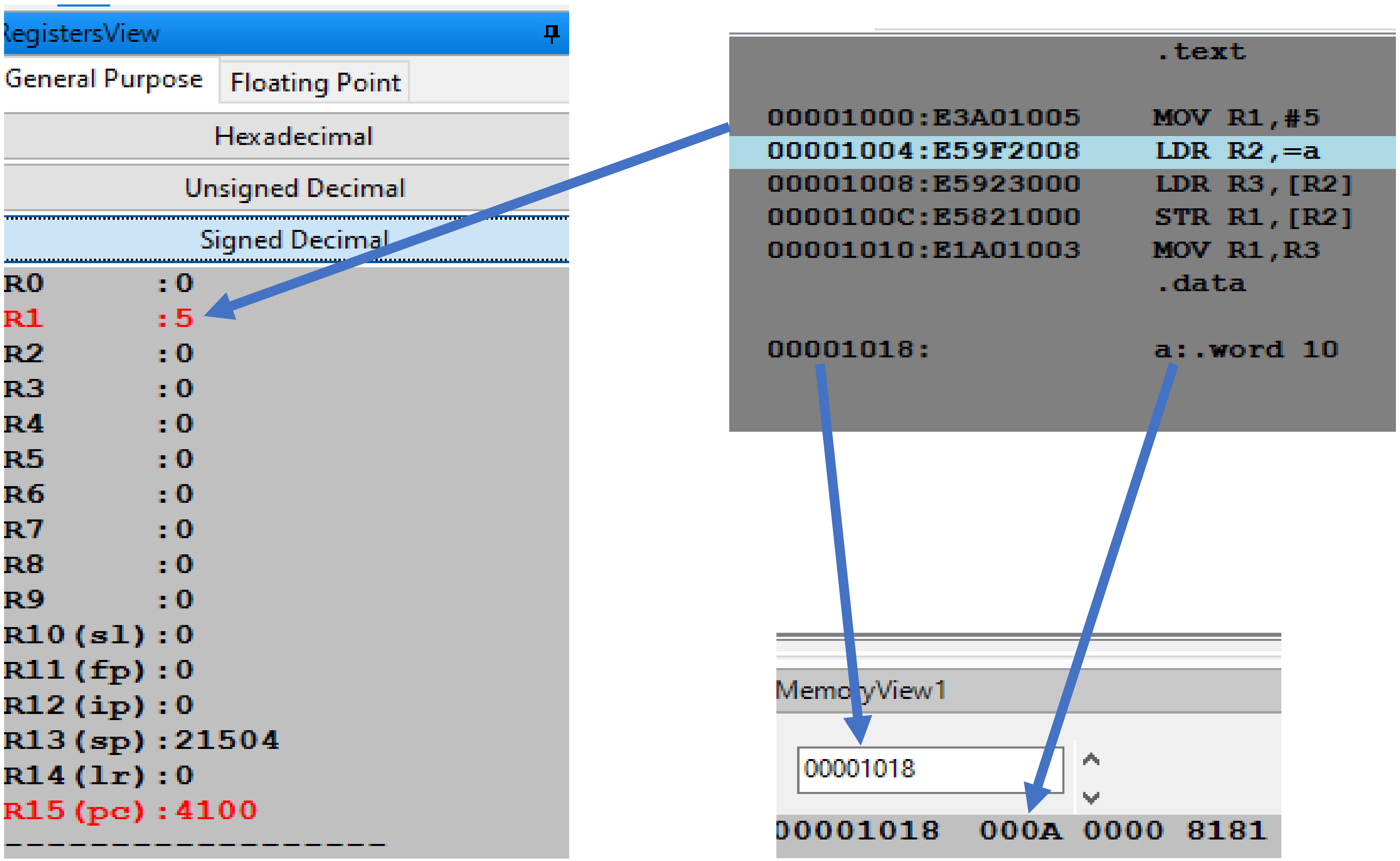


SWAP : A & R1


RegistersView
General Purpose
Floating Point
Hexadecimal
Unsigned Decimal
Signed Decimal
R0 :0
R1 :5
R2 :0
R3 :0
R4 :0
R5 :0
R6 :0
R7 :0
R8 :0
R9 :0
R10(sl):0
R11(fp):0
R12(ip):0
R13(sp):21504
R14(lr):0
R15(pc):4100
.text
00001000:E3A01005 MOV R1,#5
00001004:E59F2008 LDR R2,=a
00001008:E5923000 LDR R3,[R2]
0000100C:E5821000 STR R1,[R2]
00001010:E1A01003 MOV R1,R3
.data
00001018: a:.word 10
MemoryView1
00001018
00001018 000A 0000 8181

SWAP : A & R1

RegistersView

General Purpose | Floating Point

Hexadecimal

Unsigned Decimal

Signed Decimal

```
R0       :0
R1       :5
R2       :4120
R3       :10
R4       :0
R5       :0
R6       :0
R7       :0
R8       :0
R9       :0
R10(sl)  :0
R11(fp)  :0
R12(ip)  :0
R13(sp)  :21504
R14(lr)  :0
R15(pc)  :4112
------------------
```

```
                         .text

00001000:E3A01005        MOV R1,#5
00001004:E59F2008        LDR R2,=a
00001008:E5923000        LDR R3,[R2]
0000100C:E5821000        STR R1,[R2]
00001010:E1A01003        MOV R1,R3
                         .data

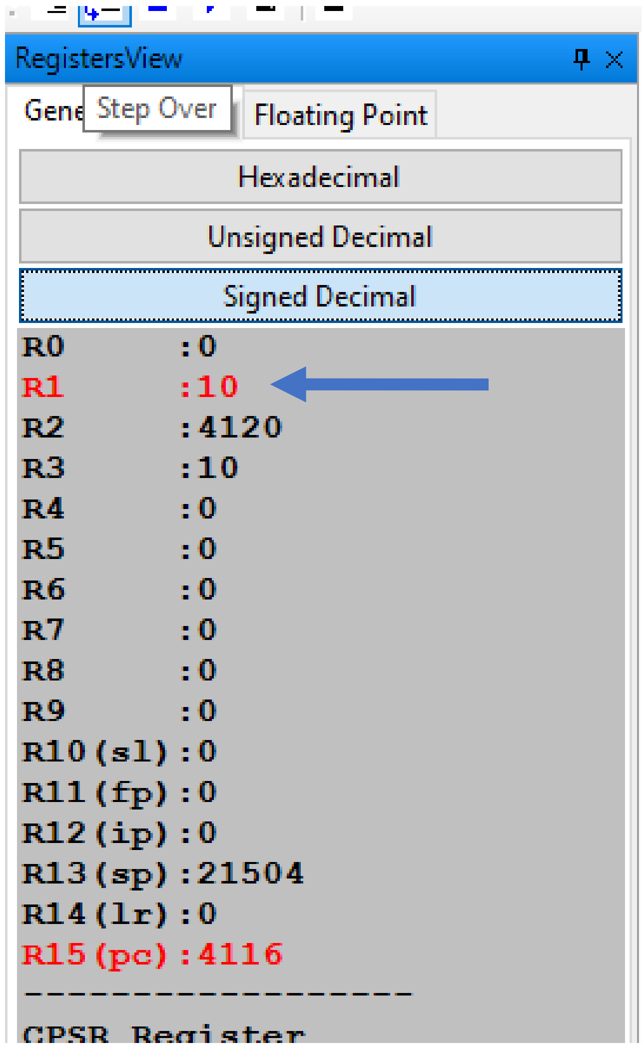
00001018:                a:.word 10
```

MemoryView1

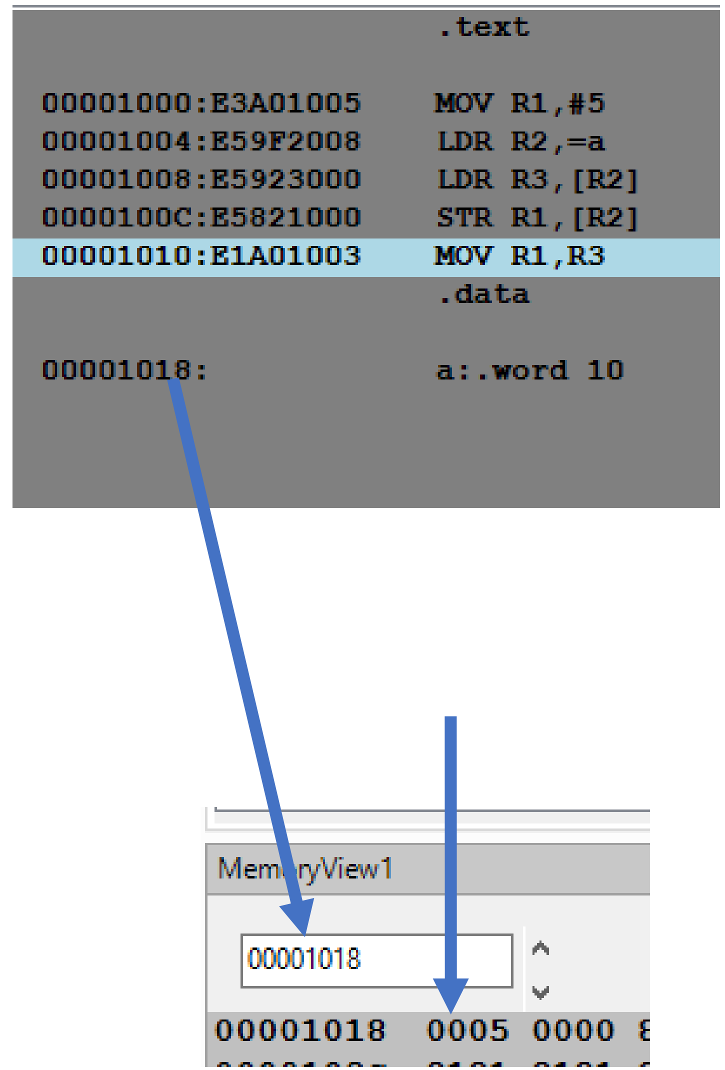
00001018

```
00001018  0005 0000 81
0000103c  8181 8181 81
```

SWAP : A & R1


RegistersView
Gene
Step Over
Floating Point
Hexadecimal
Unsigned Decimal
Signed Decimal
R0 :0
R1 :10
R2 :4120
R3 :10
R4 :0
R5 :0
R6 :0
R7 :0
R8 :0
R9 :0
R10(sl):0
R11(fp):0
R12(ip):0
R13(sp):21504
R14(lr):0
R15(pc):4116

CPSR Register


.text
00001000:E3A01005 MOV R1,#5
00001004:E59F2008 LDR R2,=a
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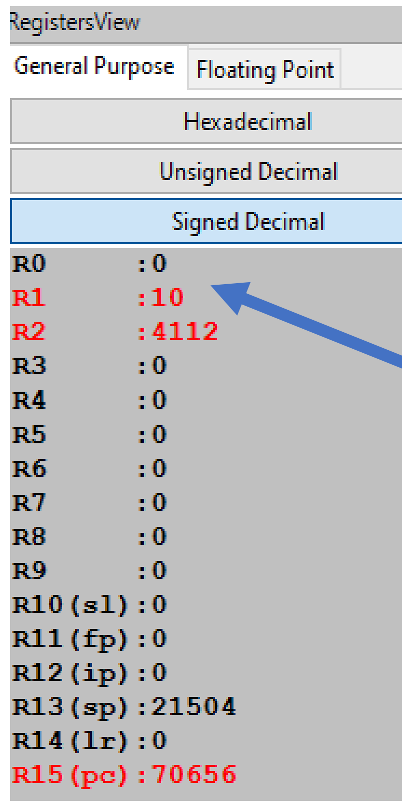
00001010:E1A01003 MOV R1,R3
.data
00001018: a:.word 10
MemoryView1
00001018
00001018 0005 0000

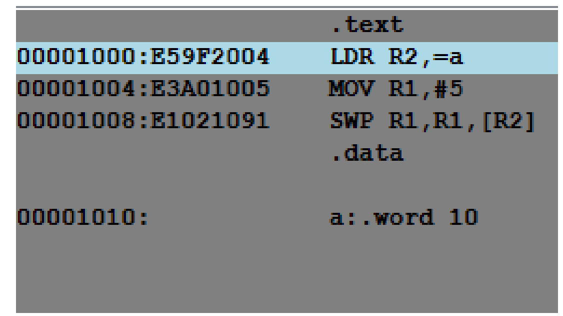
Swap Instruction

SWP <Swap Destination>, <Original>, [<address>]

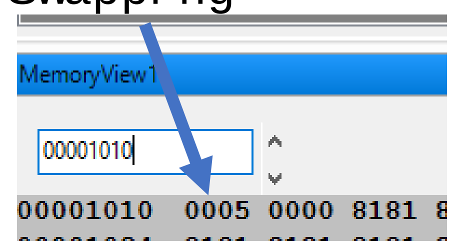
Eg: SWP R0, R1, [R2]

```
.text
LDR    R2,  =a
MOV  R1,  #5
SWP   R1,  R1,
[R2]
.data
a: .word  10
```





After Swapping



MCQ' s

1. Which instruction is used to read the contents of the Current Program Status Register (CPSR) or Saved Program Status Register (SPSR) into a general-purpose register?

A. MSR B. SWP C. MRS D. LDR

2. When using the MSR instruction to write to a status register, specific fields can be targeted. According to the slides, which bit range does the Flag Field (_f) cover?

A. Bits 0:7 B. Bits 8:15 C. Bits 16:23 D. bits24:31

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1. Which instruction is used to read the contents of the Current Program Status Register (CPSR) or Saved Program Status Register (SPSR) into a general-purpose register?

A. MSR B. SWP C. MRS D. LDR

Answer – C. MRS

2. When using the MSR instruction to write to a status register, specific fields can be targeted. According to the slides, which bit range does the Flag Field (_f) cover?

A. Bits 0:7 B. Bits 8:15 C. Bits 16:23 D. Bits 24:31

Answer – D. Bits 24:32

MCQ's

3. Consider the following instruction : SWP R1, R1, [R2]. What operation does this instruction perform?
A. It swaps the values inside registers R1 and R2.
B. It loads the word from memory at address R2 into R1 and stores the original value of R1 into memory at address R2.
C. It adds the value of R1 to the value at memory address R2.
D. It compares the value in R1 with the value at memory address R2

4. To safely modify the bits in the CPSR, What is the correct logical order of operations shown?
A. MSR (Write) → AND (Modify) → MRS (Read)
B. MRS (Read) → AND (Modify) → MSR (Write)
C. LDR (Load) → STR (Store)
D. SWP (Swap) → MRS (Read)

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D. It compares the value in R1 with the value at memory address R2

Answer - B

4. To safely modify the bits in the CPSR, What is the correct logical order of operations shown?

A. MSR (Write) → AND (Modify) → MRS (Read)

B. MRS (Read) → AND (Modify) → MSR (Write)

C. LDR (Load) → STR (Store)

D. SWP (Swap) → MRS (Read)

Answer – B. MRS (Read) → AND (Modify) → MSR (Write)

THANK YOU

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